A-A-54272
15 November 1990

COMMERCIAL ITEM DESCRIPTION

IMPRESSION MATERIAL, DENTAL, Paste, Variable Viscosity

The General Services Administration has authorized the use of this Commerical Item Description.

This commercial item description covers a universal rubber base impression material, having a gradual set increasing in viscosity from a light bodied consistency to a heavy bodied consistency, suitable for taking detailed dental impressions.

Salient characteristics:

Material. The impression material shall consist of a base paste, an accelerator paste, and a liquid tray adhesive. The base shall be a polysulfide rubber type and shall be white in color. The base may contain fillers which do not impair the serviceability of the item. The accelerator shall contain a copper salt in a neutral plasticizer fortified with fillers, and shall be blue in color. The tray adhesive shall be a fast-drying ketone solution of synthetic elastomer, and shall be dark brown in color. The solvents shall be methyl ethyl ketone, toluene, and acetone.

A free-flowing elastic material suitable for taking impressions in the mouth shall be produced when the base is mixed with the accelerator in accordance with the instructions. The minimum volume of the mixture shall be 120 milliliters (ml) when all of the base and accelerator in the package is mixed.

The adhesive shall be suitable for bonding the impression material to metal, acrylic or compound impression trays during dental operative procedures. The package shall contain a minimum weight of 1/2 ounce of adhesive.

Odor and flavor. The impression material shall have no objectionable odor or flavor.

AMSC N/A FSC 6520

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Irritation. Prior to delivery, the supplier shall furnish a statement directly to the Technical Services Branch (Code RSTS-2), Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, stating that the material does not normally irritate oral tissues and does not contain poisonous ingredients in sufficient concentration to be harmful to human beings when used as directed.

Physical properties. Physical properties of the impression material shall be in accordance with Table I.

Table I. Physical properties

Characteristic	
Mixing time (minutes, maximum)	1
Working time (minutes) (± 1 min)	5
Consistency, disk diameter (millimeters) at 12 minutes	36 to 42
Permanent deformation (percent, maximum)	3
Dimensional stability (percent, change in 24 hours, maximum)	0.4
Strain in compression (percent)	2.0 to 6.0
Rate of gain of hardness (minutes to reach a durometer reading of 18)	15 ± 1
Flow (percent, maximum)	0.36

Procedures for the consistency, permanent deformation (compression set), dimensional stability, strain in compression, and flow tests shall be as specified in American Dental Association (ADA) Specification No. 19 for Non-Aqueous Elastomeric Dental Impression Materials. Procedures for the mixing time, working time and hardness tests shall be as specified below.

Impression material shall comply with the requirements for reproduction of detail and compatibility with gypsum as specified in ADA Specification No. 19. The line width shall be 0.020 millimeters (mm).

Impression material shall be compatible with both acid copper plating baths and alkaline silver plating baths when tested as specified in ADA Specification No. 19.

Impression material shall be tested for deterioration as specified below.

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Instructions for use. Adequate and accurate instructions for manipulating the material shall accompany each unit package. These instructions shall include the following minimum requirements:

1. The proportioning of base and accelerating materials to give a proper mix by weight and extruded length.

2. The mixing time required to produce a streak-free mix.

3. Working time and minimum time for leaving impression in the mouth at normal room temperature.

4. Precautionary statement on finger (or skin) and clothing contact, and removal of stains.

5. Precautionary statement on storage in cool place.

Certification. At the time of award, suppliers shall furnish a dated certificate and data showing that the brand of impression material they propose to furnish is the same in quality and brand as that appearing on the list of certified materials for American Dental Association Specification No. 19. Compliance with this certification requirement does not absolve the contractor from complete compliance with the requirements of this Commercial Item Description.

Delivery schedule. Not more than 3 months shall have elapsed from the date of manufacture of the impression material to the date of delivery to the Government.

Immediate container. The base paste and accelerator paste shall be supplied in airtight, squeezable, metal composition tubes fabricated of material that will not react with nor contaminate the impression material. The tubes shall be of sufficient strength and flexibility to prevent leakage and shall have screwcaps.

The immediate container for the liquid adhesive shall be a glass bottle with screwcap. An applicating brush shall be permanently affixed to the underside of the screwcap. The bottle of adhesive shall not leak during ordinary handling, shipping and storage.

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Marking. Each tube and unit package of material shall be marked with the lot or control number and date of manufacture. In addition, each tube of polysulfide rubber base shall be permanently and legibly identified as "BASE"; each tube of accelerator, as "ACCELERATOR"; and each bottle of adhesive, as "ADHESIVE."

Workmanship. The base and the accelerator shall be homogeneous, uniform in consistency and free of defects, foreign materials, and impurities which detract from their appearance or which may impair their serviceability.

Unit. Package (PG). One package containing one tube of base paste, one tube of accelerator paste, one bottle of adhesive, and instructions for use constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one package of impression material. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, (percent defective) and sampling for tests shall be conducted at inspection level S-1, AQL 1.0 (percent defective).

Examination. The impression material shall be examined for defects including, but not limited to, those listed below.

Defects

Package of impression material not complete with base, accelerator, and adhesive.
Immediate containers not proper type, as specified.
Instructions for use missing, incomplete, or illegible.
Container for adhesive doesn't have applicator brush permanently affixed to underside of cap.
Tubes and bottle and unit package not marked as specified.
Components of impression material have defects which detract from their appearance or impair their serviceability.
Components of impression material not proper color, as specified.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Standard testing conditions. All physical tests shall be made under uniform atmospheric conditions 23.0 ± 1.0°C. (73.4 ± 1.8°F.) and 50 ± 5 percent relative humidity, unless otherwise specified. Equipment and material shall be conditioned in the testing room for not less than 10 hours prior to making the tests.

Standard mixes. The size of the standard mix to be used depends on the size of the test specimen to be made. A small mix shall consist of amounts of base and accelerator to give 5 ± 0.5 grams (gm.) of the mixed material. The large mix shall consist of amounts of base and accelerator to give 20 ± 1 gm. of the mixed material.

Mixing time. Impression material shall be spatulated in accordance with furnished instructions. The time to produce a homogeneous, smooth mix, free from lumps, granules, and color streaks shall be recorded.

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Working time. A small standard mix shall be prepared and retained in a compact layer approximately 3 mm. thick. A 10 gm. steel rod, 2.5 mm. in diameter (see figure 1), shall be applied to the surface at 20 second intervals and raised slowly. The needle shall be withdrawn from the test material at the same rate at which the needle penetrated the material. The end of the rod shall be wiped clean after each application. The working time shall be recorded as the time elapsed from the start of the mix to the time at which the material fails to string and an indentation persists when the rod is withdrawn. Three determinations shall be made and the average working time shall be reported to the nearest 1/4 minute.

Rate of gain of hardness. A large standard mix of impression material shall be prepared and retained in an overfilled Durometer cup (see figure 2). The overfill shall be from 1/16 to 1/8 inch thick. The top of the specimen shall be formed flat and parallel to the plate.

The hardness characteristics of the impression material shall be tested using a Shore Type A Durometer, or equivalent testing equipment. Hardness readings shall be made at one minute intervals starting as soon as the rubber is sufficiently tack-free to permit testing. The hardness of the impression material is determined by pressing the durometer instrument firmly against the set rubber. Care must be taken to keep the indentor point at least 1/4-inch from the side walls of the cup. The pressure of the instrument against the sample must be applied quickly, and the maximum needle reading noted. The semi-cured rubbers permit considerable needle drift, if a time interval exists, between application of pressure and reading the instrument. The application of pressure and reading the instrument must occur simultaneously. Three readings shall be averaged for each unit of product tested. Defective readings shall be discarded and additional readings taken. A new spot shall be penetrated for each reading. The foot of the instrument shall be wiped clean before each penetration. The total elapsed time from the start of the mix to reach the required hardness shall be reported to the nearest minute.

Deterioration. Tubes of accelerator and base paste shall be stored at 60°C ± 2°C for 7 days. The tubes shall show no evidence of leakage after storage. The permanent deformation (compression set) test shall be performed on this material. The permanent deformation shall not exceed 4.0 percent.

Adhesive performance. The adhesive shall properly bond the impression material to impression trays when used according to the manufacturer's instructions.

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Leakage (adhesive bottle). The filled container shall be suspended in an inverted position for six hours at room temperature 70°F ± 3°F.). No leakage or loss of weight of contents shall occur during or at the completion of the test.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pounds units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and of regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.
NOTE: The following National Stock Number is covered by this document.

National Stock Number	Item Identification
6520-00-118-1167	IMPRESSION MATERIAL, DENTAL, Paste, Variable Viscosity

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

AGENT:

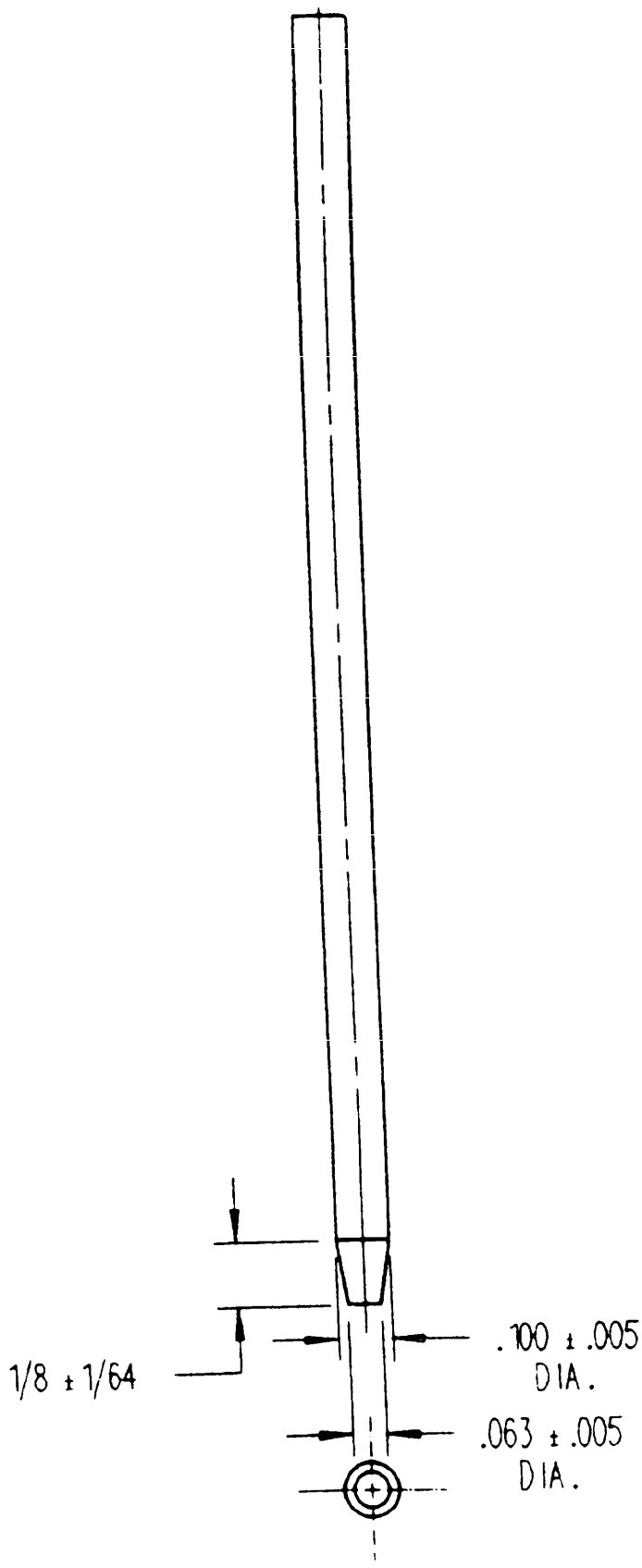
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CIVIL AGENCY COORDINATING ACTIVITIES:
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PROJECT NO. 6520-2355
LOCATION: NCR - DISK 8

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ADJUST LENGTH TO
GIVE 10 gm. WEIGHT

NOTE:

A SUITABLE NON-RESTRAINING GUIDE MAY BE USED WITH THE NEEDLE.

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.

FIGURE 1. WORKING TIME TEST NEEDLE

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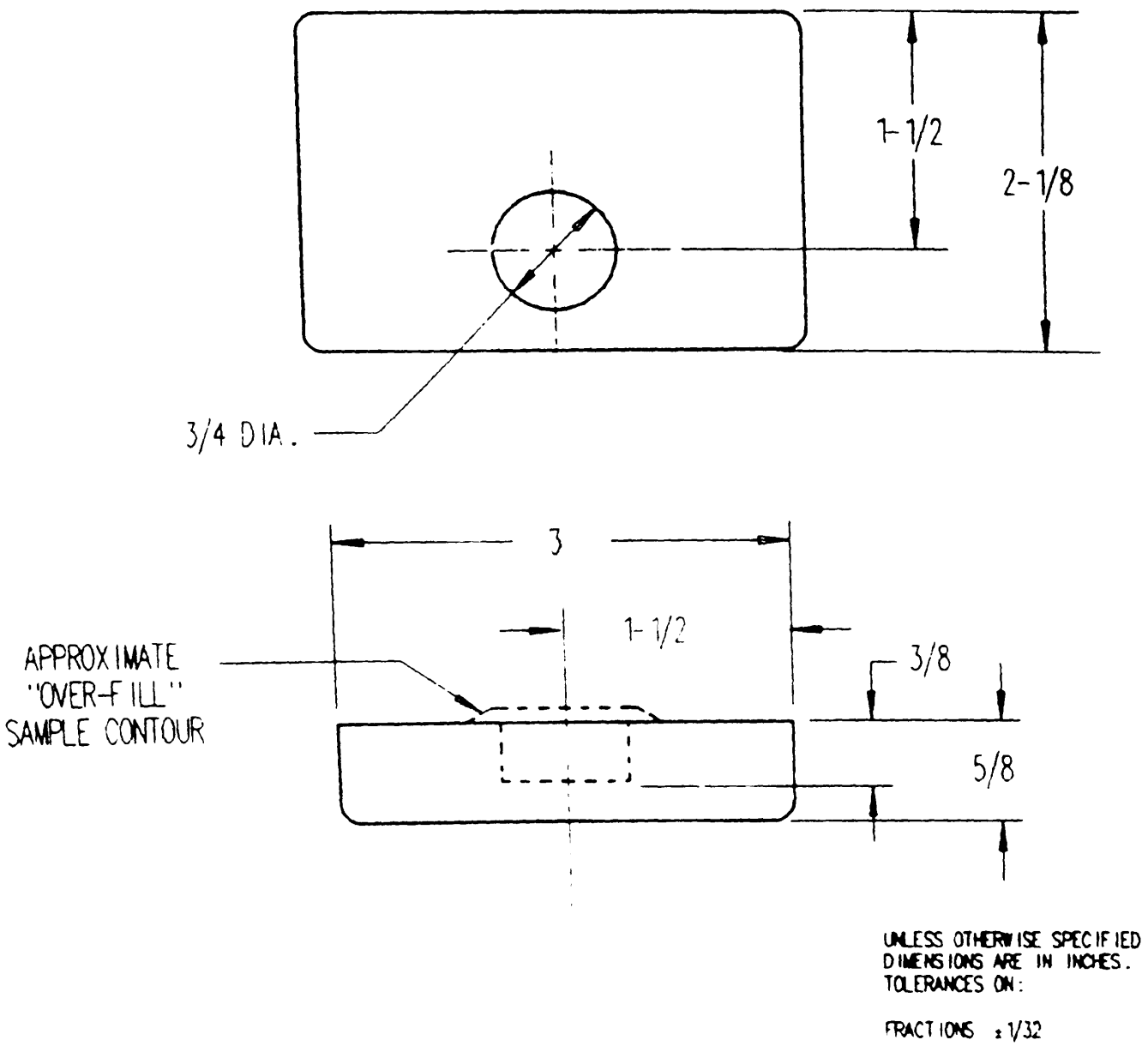


FIGURE 2. DUROMETER CUP FOR HARDNESS TEST

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-54272	IMPRESSION MATERIAL, DENTAL, Paste, Variable Viscosity

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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